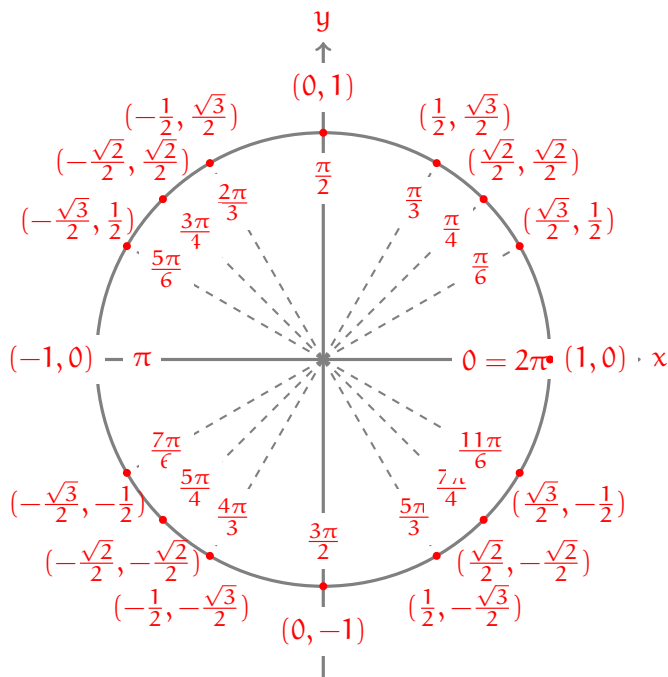


**Integral Formulas.**

- $\int \frac{1}{1+t^2} dt = \arctan t + C$
- $\int \frac{1}{\sqrt{1-t^2}} dt = \arcsin t + C$
- $\int \frac{1}{t} dt = \ln |t| + C$

Trig Identities.

- power reduction: $\cos^2 \theta = \frac{1}{2} + \frac{1}{2} \cos 2\theta$
 $\sin^2 \theta = \frac{1}{2} - \frac{1}{2} \cos 2\theta$
- double angle: $\sin 2\theta = 2 \sin \theta \cos \theta$
 $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$

Dot and Cross Products. Let $\vec{v}$, $\vec{w}$, $\vec{r}$ be vectors.

- $\vec{v} \cdot \vec{w} = \|\vec{v}\| \|\vec{w}\| \cos \theta$
- $\|\vec{v} \times \vec{w}\| = \|\vec{v}\| \|\vec{w}\| \sin \theta = [\text{parallelogram area}]$
- $|\vec{v} \cdot (\vec{w} \times \vec{r})| = [\text{parallelepiped volume}]$

Orthogonal Projections. Let $\vec{v} \neq \vec{0}$ and $\vec{w}$ be vectors.

- $\text{proj}_{\vec{v}}(\vec{w}) = \left(\frac{\vec{v} \cdot \vec{w}}{\vec{v} \cdot \vec{v}} \right) \vec{v}$
- $\text{comp}_{\vec{v}}(\vec{w}) = \frac{\vec{v} \cdot \vec{w}}{\|\vec{v}\|}$

Distance Formulas. Let Q be a point, $\mathcal{P}$ a plane with normal $\vec{n}$, and ℓ a line with direction vector $\vec{v}$.

- $\text{distance}(Q, \mathcal{P}) = \frac{|\vec{PQ} \cdot \vec{n}|}{\|\vec{n}\|}$ where P is a point on $\mathcal{P}$
- $\text{distance}(Q, \ell) = \frac{\|\vec{PQ} \times \vec{v}\|}{\|\vec{v}\|}$ where P is a point on ℓ

Quadratic Surfaces. Standard equations.

- paraboloid: $z = x^2 + y^2$
- saddle: $z = x^2 - y^2$
- 1-sheeted hyperboloid: $x^2 + y^2 - z^2 = 1$
- 2-sheeted hyperboloid: $-x^2 - y^2 + z^2 = 1$
- ellipsoid—sphere if uniformly scaled:
 $x^2 + y^2 + z^2 = 1$
- double-cone: $z^2 = x^2 + y^2$

Tangent Planes. Let $f(x, y)$ be a function of two variables, $F(x, y, z)$ be a function of three variables, and C be a constant.

- tangent plane to the graph $z = f(x, y)$ at $P(a, b, \star)$ is:
 $z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$
- tangent plane to level set $F(x, y, z) = C$ at point P is:
 $\vec{\nabla} F(P) \cdot \vec{x} = \vec{\nabla} F(P) \cdot \vec{p}$

Directional Derivatives. Let f be a function and $\vec{u}$ be a unit direction in its domain.

- $D_{\vec{u}} f(P) = \vec{\nabla} f(P) \cdot \vec{u} = \|\vec{\nabla} f(P)\| \cos \theta$